



॥वसुधैव कुटुम्बकम्॥

SYMBIOSIS

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ASSIGNMENT No: 12

TITLE: Write a Java program to create and use user-defined packages.

Problem Statement : Write a Java program to create and use user-defined packages.

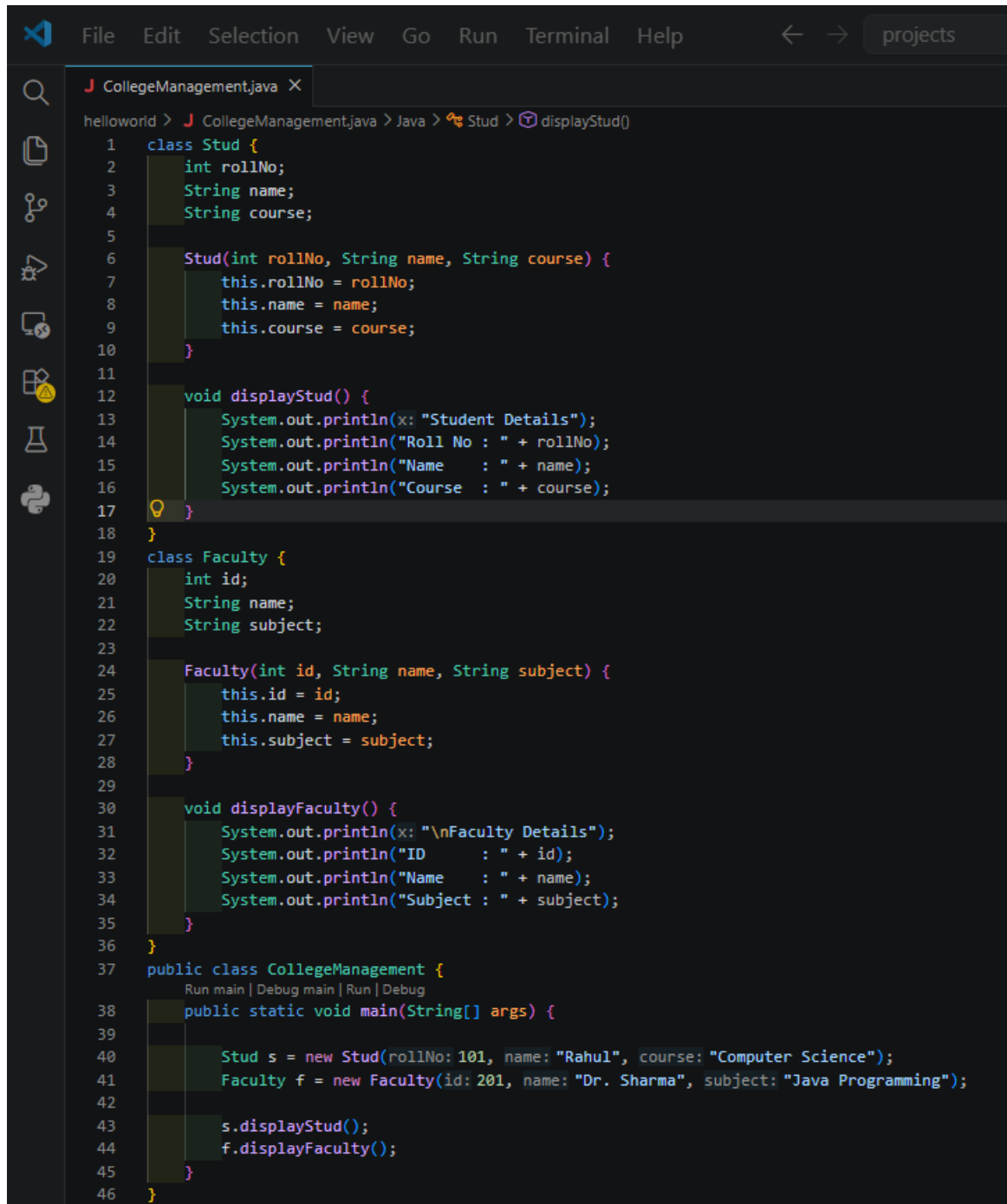
Learning Objectives : To create and use user-defined packages for organizing Java programs.

Theory: Packages are used to organize related classes and interfaces into namespaces. User-defined packages improve code organization, reduce naming conflicts, and enhance reusability. They also support access protection and modular application development. Java packages simplify project management by grouping logically related components together. Largescale software systems extensively use packages.

Exercise:

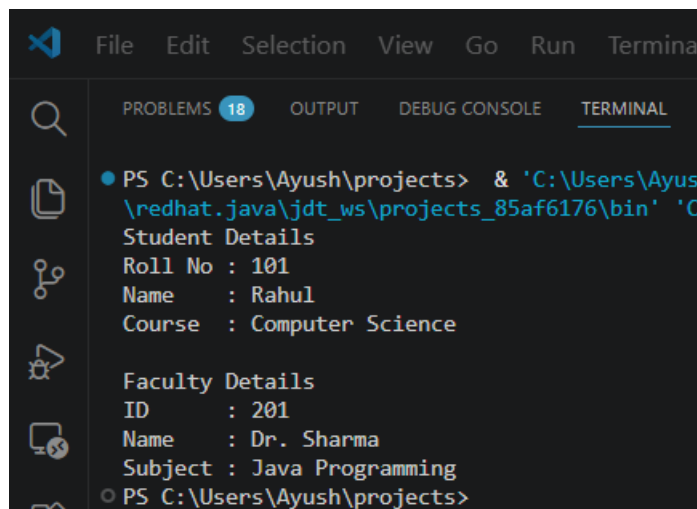
1. Create a Java program by developing user-defined packages for a College Management System. Create separate packages for student and faculty details, and display their information using classes from these packages.

Program:



```
File Edit Selection View Go Run Terminal Help
CollegeManagement.java X
helloworld > J CollegeManagement.java > Java > Stud > displayStud()
1 class Stud {
2     int rollNo;
3     String name;
4     String course;
5
6     Stud(int rollNo, String name, String course) {
7         this.rollNo = rollNo;
8         this.name = name;
9         this.course = course;
10    }
11
12    void displayStud() {
13        System.out.println(x: "Student Details");
14        System.out.println("Roll No : " + rollNo);
15        System.out.println("Name : " + name);
16        System.out.println("Course : " + course);
17    }
18 }
19 class Faculty {
20     int id;
21     String name;
22     String subject;
23
24     Faculty(int id, String name, String subject) {
25         this.id = id;
26         this.name = name;
27         this.subject = subject;
28     }
29
30     void displayFaculty() {
31         System.out.println(x: "\nFaculty Details");
32         System.out.println("ID : " + id);
33         System.out.println("Name : " + name);
34         System.out.println("Subject : " + subject);
35     }
36 }
37 public class CollegeManagement {
38     Run main | Debug main | Run | Debug
39     public static void main(String[] args) {
40
41         Stud s = new Stud(rollNo: 101, name: "Rahul", course: "Computer Science");
42         Faculty f = new Faculty(id: 201, name: "Dr. Sharma", subject: "Java Programming");
43
44         s.displayStud();
45         f.displayFaculty();
46     }
47 }
```

Output:

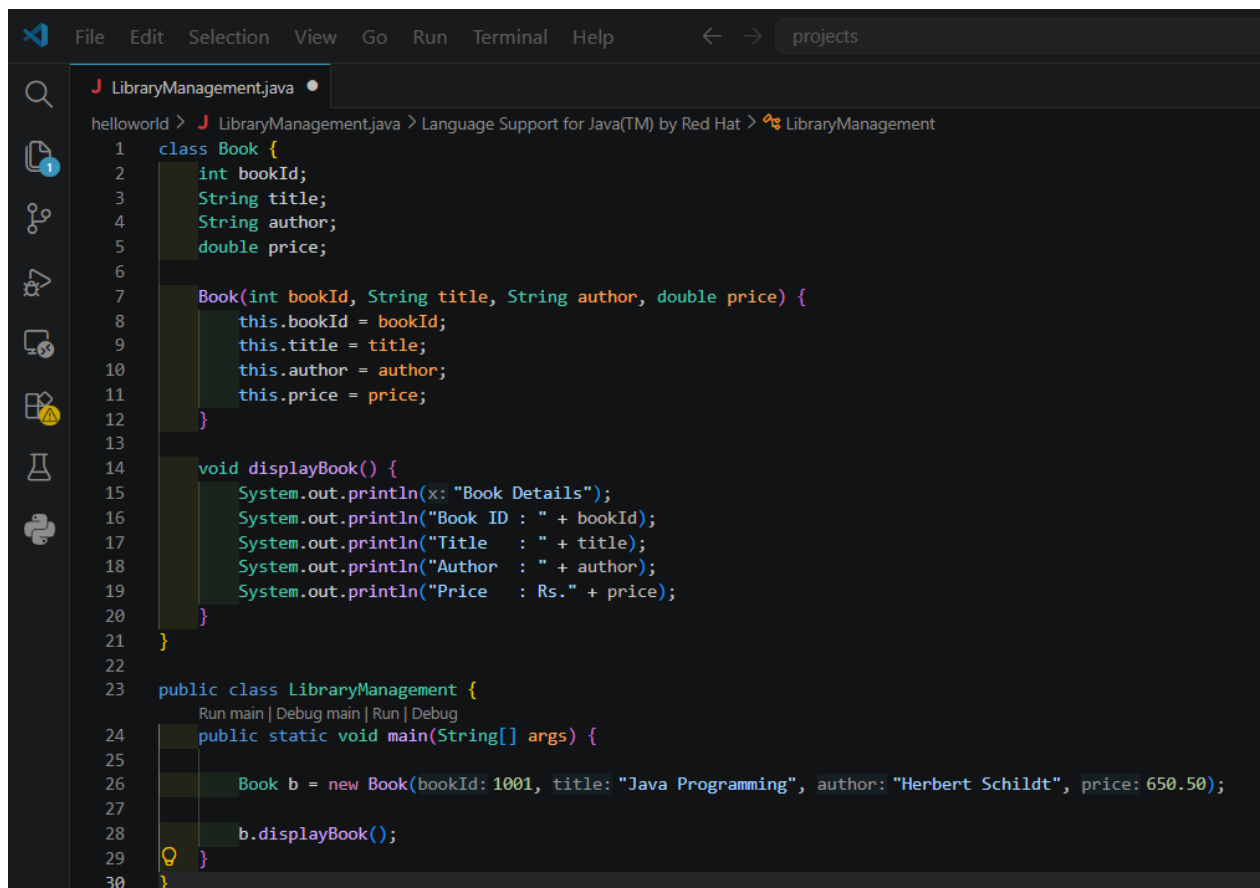


```
PS C:\Users\Ayush\projects> & 'C:\Users\Ayush\redhat.java\jdt_ws\projects_85af6176\bin' 'C:\Users\Ayush\projects'
Student Details
Roll No : 101
Name : Rahul
Course : Computer Science

Faculty Details
ID : 201
Name : Dr. Sharma
Subject : Java Programming
PS C:\Users\Ayush\projects>
```

2. Create a Java program by developing a user-defined package named library that contains a class to store and display book details such as book ID, title, author, and price. Import the package into the main program and display the book information.

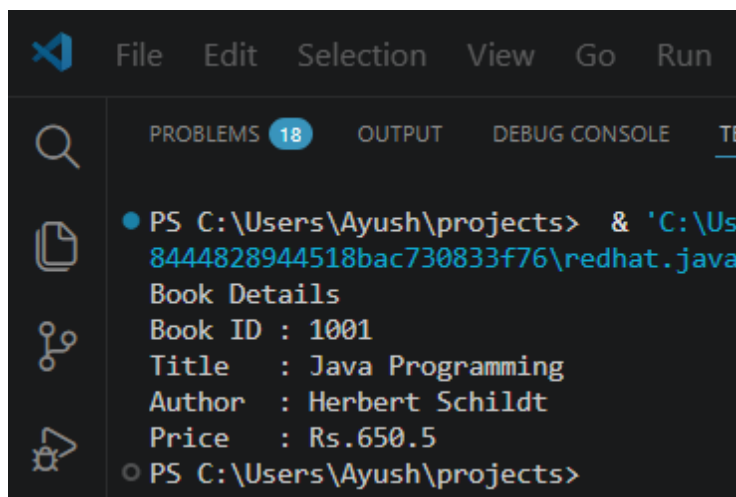
Program:



```
File Edit Selection View Go Run Terminal Help
LibraryManagement.java
helloworld > J LibraryManagement.java > Language Support for Java(TM) by Red Hat > LibraryManagement

1 class Book {
2     int bookId;
3     String title;
4     String author;
5     double price;
6
7     Book(int bookId, String title, String author, double price) {
8         this.bookId = bookId;
9         this.title = title;
10        this.author = author;
11        this.price = price;
12    }
13
14    void displayBook() {
15        System.out.println("Book Details");
16        System.out.println("Book ID : " + bookId);
17        System.out.println("Title : " + title);
18        System.out.println("Author : " + author);
19        System.out.println("Price : Rs." + price);
20    }
21 }
22
23 public class LibraryManagement {
24     Run main | Debug main | Run | Debug
25     public static void main(String[] args) {
26
27         Book b = new Book(bookId: 1001, title: "Java Programming", author: "Herbert Schildt", price: 650.50);
28
29         b.displayBook();
30     }
31 }
```

Output:

A screenshot of the Visual Studio Code interface. The top menu bar shows 'File', 'Edit', 'Selection', 'View', 'Go', and 'Run'. Below the menu bar, the 'PROBLEMS' tab is active, showing 18 items. The terminal window displays the following output:

```
PS C:\Users\Ayush\projects> & 'C:\Us  
8444828944518bac730833f76\redhat.java  
Book Details  
Book ID : 1001  
Title : Java Programming  
Author : Herbert Schildt  
Price : Rs.650.5  
PS C:\Users\Ayush\projects>
```